GEK-83832

*Instructions*

# Aydin Monitor
# 8830 Analog Patriot Series
# High Resolution, 19-Inch
# Color Monitor
# Installation and Maintenance

# AYDIN MONITOR

## 8830 ANALOG PATRIOT SERIES HIGH RESOLUTION, 19-INCH COLOR MONITOR

## INSTALLATION AND MAINTENANCE

GEK-83832

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## NOTICE

This manual is an edited reproduction of selected portions of the Aydin Controls 8830 Analog Patriot Series High Resolution, 19-Inch Color Monitor Operation and Maintenance Manual, document no. 150-8830-001. Complete copies of the Aydin manual can be obtained from:

AYDIN CONTROLS
414 Commerce Drive
Fort Washington, PA 19034

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, 24153.

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# SECTION 1

# GENERAL DESCRIPTION

## 1.1 INTRODUCTION

This manual contains installation, operation, and maintenance information for the AYDIN CONTROLS 8830 Analog Patriot Series Color Monitor. The Monitor is an in-line gun, high resolution red, green, blue (RGB), shadow mask display device designed for compatibility with alphanumeric and graphic raster-scan display systems. It features a fine pitch shadow mask CRT and separate red, green, and blue video channels to meet high resolution display requirements. The unit also contains provisions for external synchronization. Basic operational features are compatible with AYDIN CONTROLS raster-scan display systems.

To ensure compatibility with standard industrial TV design techniques, the monitor meets the specifications of EIA Standard RS-170. It also complies with DHEW Rules and Regulations, Part 21CFR Subchapter J, relating to x-ray radiation.

## 1.2 FUNCTIONAL DESCRIPTION

The 8830 Analog Color Monitor produces a raster scan display using basic television receiver design principles. The monitor accepts external sync and separate red, green, and blue video signal inputs formatted in accordance with EIA Standard RS-170. For each color, separate video inputs allow wide color bandwidth, which, together with a high resolution shadow mask CRT and inline gun, enables the monitor to produce a high resolution color display with stable fixed convergence.

A three-gun CRT with a fine pitch shadow mask is complemented by a precision gun assembly to produce small spot sizes. Pincushion correction circuits correct magnetic deflection, minimizing geometric distortion. Regulated low voltage and high voltage dc power ensures stable display characteristics.

To display alphanumeric and graphic data generated by a display generator, digital data developed by a computer is converted into properly formatted red, green, and blue video. Synchronization signals are included with the green video signal or as separate external sync.

Computer data is generally supplied to a display generator, which converts digitally-coded computer data into a dot matrix field pattern for display compatibility. The digital dot matrix patterns are then converted into composite video signals.

## 1.3 PHYSICAL DESCRIPTION

The 8830 Analog Color Monitor includes a finished cabinet enclosure designed for table-top application. For rack-mounted applications, a slide kit that mounts directly to the chassis is supplied with the monitor. Mounting holes and hardware are provided for slide mounting.

Figure l-l lists and depicts the location of the monitor's printed circuit boards (PCBs or cards) and major components. Section 4 explains and provides the location of maintenance adjustments.

## 1.4 CHARACTERISTICS

Table l-l lists the 8830 Analog Color Monitor's essential electrical performance and physical characteristics.

## 1.5 RECOMMENDED TEST EQUIPMENT AND SPECIAL TOOLS

The following test equipment and special tools, or their equivalents, are recommended for maintaining the 8830 Analog Color Monitor:

a. Digital Multimeter, Weston Model 1240, or equivalent

b. High Voltage Probe, 30kV - Simpson, Catalog No. 00509

c. Oscilloscope, Hewlett-Packard, Type 184A, 2-Channel, 50MHz; Amplifier, Type 1801-A; Time Base and Delay Generator, Type 1821A; Two 10-l probes, 10004D; and a 50-to-l probe, 10002A.

d. Degaussing Coil, G. C. Electronics, Catalog No. 9317.

e. Alignment Screwdriver, G. C. Electronics, Catalog No. 5009.

f. Simpson Model 260 multimeter (use with item b).

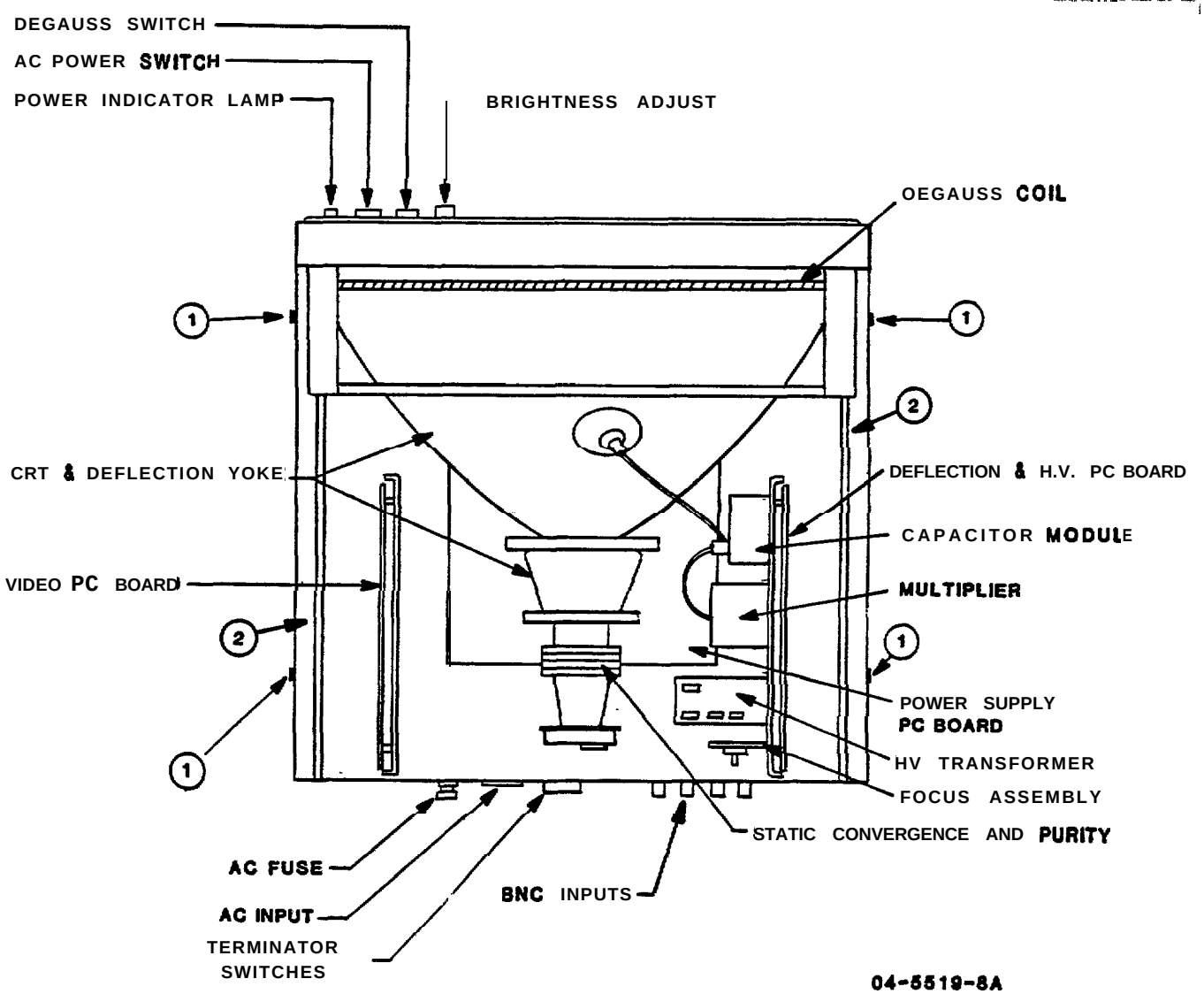



Figure 1-1
19-INCH ANALOG COLOR MONITOR, MODEL 8830
(TOP VIEW, WITHOUT CABINET)

Table l-l
8830 ANALOG COLOR MONITOR CHARACTERISTICS

| CHARACTERISTICS | DESCRIPTION |
|---|---|
| **ELECTRICAL** | |
| • Video (RS 170 compatible) | Composite: Loop-through or switchable to 75 ohms termination; 1.0V P-P nominal (0.35V to 2.0V); sync negative<br><br>Noncomposite: Loop-through or switchable to 75 ohms termination; 0.7V P-P nominal (0.25V to 1.4V); black negative |
| • Data Input | Separate inputs: red, green, and blue; composite on green for synchronization or external sync |
| • External sync | Mixed sync: Loop-through or switchable to 75 ohms termination; 2.0V P-P nominal (1.0V to 4.0V); sync negative |
| • Pulse response | Rise and Fall: Less than 20 ns |
| • Input connector | BNC type |
| • Video bandwidth | ±3 db, 50 Hz to 25 MHz for each input: red, green, and blue |
| • Black level stability | Pedestal clamp circuits; black level is maintained within 1% from APL of 10% to 90% |
| • Horizontal frequency | Available in 3 ranges: 15.0 to 18.5 kHz<br>18.5 to 21.5 kHz<br>21.5 to 25.0 kHz |
| • Horizontal retrace time | Typically 7.5 usec |
| • Vertical retrace time | Typically 650 usec |
| • Input Power | Ac 115V +10% or 220v ±10%, 50/60 Hz; operating -voltage selected by internal connector change |
| • Power consumption | 80 watts (nominal) |
| • Electrical components | 100% solid-state with the exception of CRT |
| • Display area | Effective display area 14.96 in. wide by 11.02 in. high |
| • ❀ | 19-inch; self-convergence, high resolution fine pitch shadow mask with in-line gun |

Table l-l (continued)
**8830 ANALOG COLOR MONITOR CHARACTERISTICS**

| CHARACTERISTICS | DESCRIPTION |
|---|---|
| **ENVIRONMENTAL** | |
| • Operational | Temperature: 0°C to +50°C free air at electronics<br>Rel. Humid.: 0 to 95% (noncondensing)<br>Altitude: 0 to 10,000 ft |
| • Non-operational | Temperature: -30°C to +60°C<br>Rel. Humid.: 0 to 95%<br>Altitude: 0 to 45,000 ft |
| **PERFORMANCE** | |
| • Convergence | Does not deviate more than .70 mm (.0275 in) in a centrally located area bound by the circle with a diameter equal to the picture height. Elsewhere does not deviate more than 1.2 mm (.0472 in) |
| • Linearity and geometry | No point on the raster deviates from its proper position by more than 2% of raster height |
| • Raster regulation | Maximum raster deviation is less than 2 mm (.0787 in) with a beam current change of 0 to 450 uA |
| • Resolution | Greater than 4000 characters<br>Greater than 700 pixels/line |
| MECHANICAL | |
| • Size - cabinet model | Height: 17.56 in.<br>Width: 19.25 in.<br>Depth: 21.25 in. top, 19.25 in. bottom |
| • Weight | 60 lb. without a cabinet |
| • Construction | All metal chassis with glass epoxy PCBs |

Table 1-1 (continued)
8830 ANALOG COLOR MONITOR CHARACTERISTICS

| CHARACTERISTICS | DESCRIPTION |
|---|---|
| CONTROLS | |
| • Front panel | - Brightness (potentiometer)<br>- Power (ON/OFF)<br>- Degauss (momentary switch) |
| • Internal | |
| PCB video | - Red gain<br>- Green gain<br>- Blue gain<br>- Horizontal hold<br>- Horizontal phase<br>- Red bias<br>- Green bias<br>- Blue bias<br>- Int sync (S1) |
| PCB deflection | - Vertical hold<br>- Vertical height<br>- Vertical linearity<br>- Vertical centering<br>- Width<br>- Horizontal centering<br>- Side pincushion<br>- G1<br>- G2 |
| • Rear panel | - Red input: 2 coaxial connectors<br>- Green input: 2 coaxial connectors<br>- Blue input: 2 coaxial connectors<br>- Ext. sync: 2 coaxial connectors<br>- Red: 75 ohm termination switch<br>- Green: 75 ohm termination switch<br>- Blue: 75 ohm termination switch<br>- Ext synch: 75 ohm termination switch<br>- Power connector<br>- Focus |

# SECTION 2

# INSTALLATION

## 2.1 GENERAL

This section contains procedures for installing and verifying the basic operation of the 8830 Analog Color Monitor. The monitor is thoroughly adjusted and checked-out at the factory, but may require certain minor adjustments to adapt it to a particular display generator or controller, and to compensate for minor adjustment disturbances caused during shipment. Section 4 describes complete adjustment procedures and other basic checks; however, initial installation should require only selected, simple procedures.

## 2.2 UNPACKING

The 8830 Analog Color Monitor is shipped completely assembled. It is normally packaged in a separate shipping container, unless it is incorporated in a system by AYDIN CONTROLS. Unpack the monitor as follows:

1. Carefully open the top of the container.
2. Remove the inside packing material and lift out the monitor.

## 2.3 MOUNTING INSTRUCTIONS

The cabinet-style monitor is designed to sit on a supporting surface within the user's system configuration. To remove the cabinet, remove four screws located on the bottom right and left sides.

### 2.3.1 Rack Mounting

The rack slide configuration should be installed as follows:

1. Remove the case screws described in section 2.3.
2. Remove the 2 case-mounting brackets. Each bracket contains 3 screws. This will expose three No. 8-32 screw holes.
3. Mount each slide to the chassis with three No. 8-32 x 3/8-inch screws.
4. Mate the chassis slides on the monitor with the slide receptacles in the cabinet, then slide the monitor into position.

**NOTE**

The Model 8830 Analog Color Monitor is designed for slide mounting in a standard 19-inch equipment rack. The rack kit contains the necessary hardware.

## 2.4 VIDEO INPUTS

Video amplifiers in the 8830 Analog Color Monitor are linear amplifiers that drive the CRT cathodes. The red, green, and blue video cables should be connected to the appropriate video input connectors located on the rear connector panel. (Refer to Table 2-1 and Figure l-1.) The video signals should be compatible with the timing requirements of EIA-STD-RS-170. Figure 2-1 shows the basic horizontal and vertical timing.

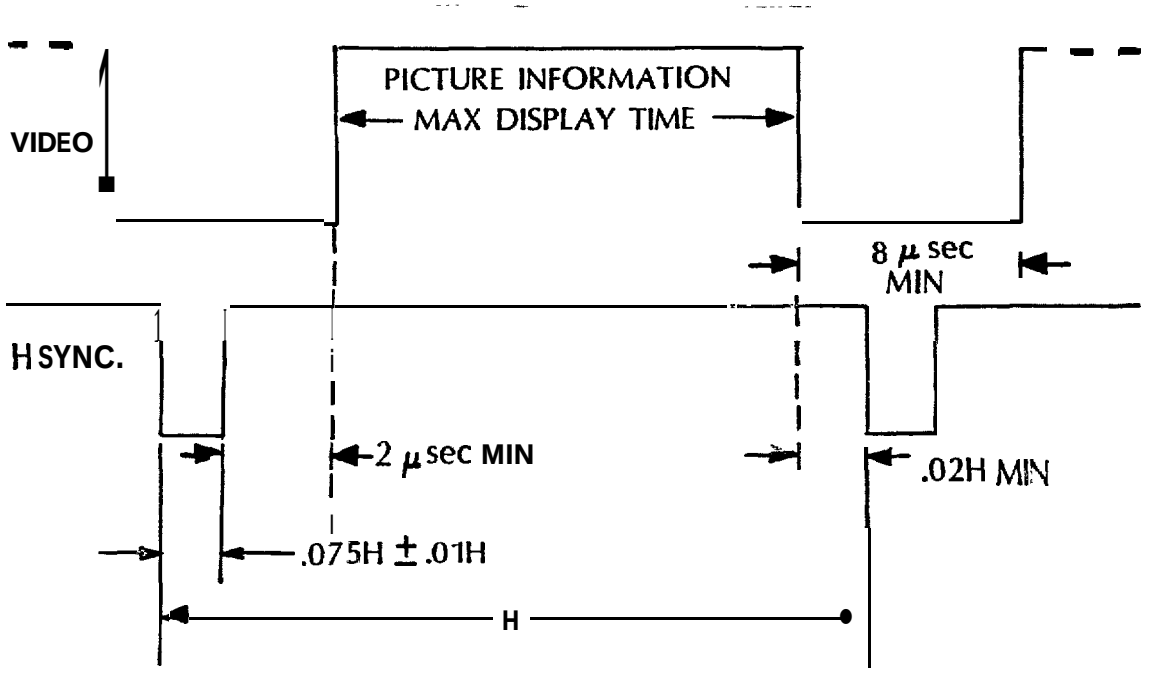
**NOTE**

The video cables should be constructed with 75 ohm coaxial cable (type RG59/U or equivalent), and terminated at the monitor cable end with the terminator switches provided. No separate sync input is required. The green video signal input to the monitor must contain composite sync. An external sync connector is provided for external sync control, if desired. The external sync switch S1 must be in its proper position.

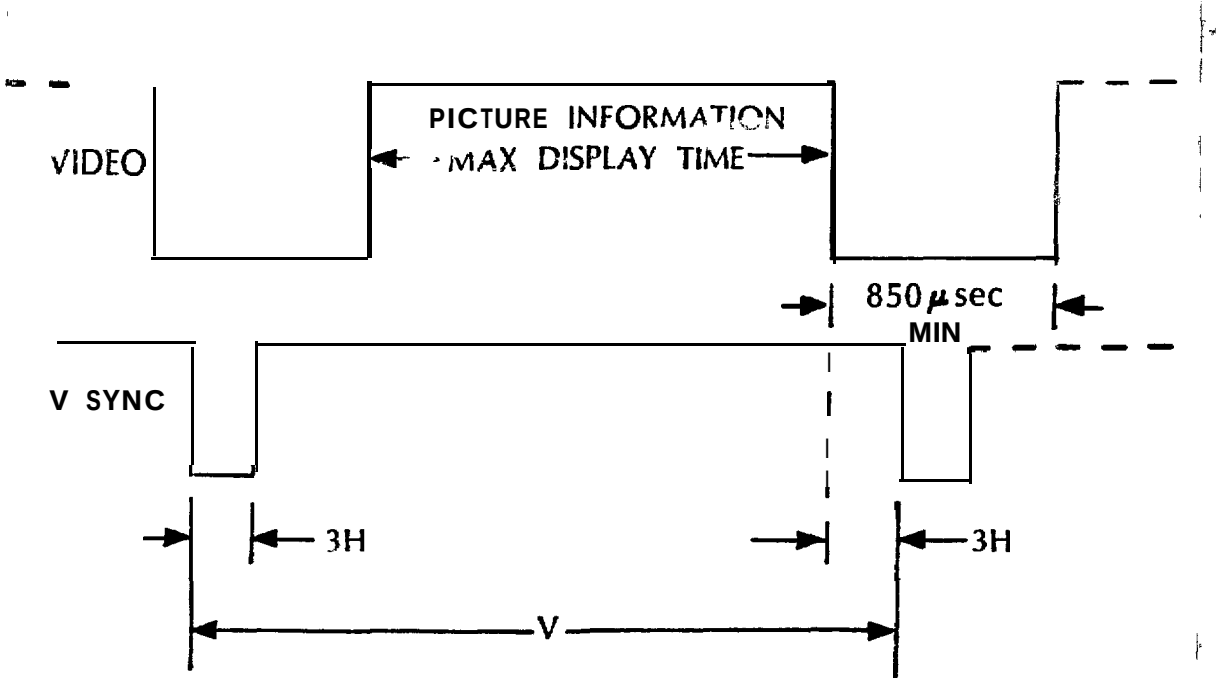
The monitor contains two connectors for each video input in order to provide for loop-through operation. In addition, the input impedance of each input can be set to a high impedance or to 75 ohms by means of three input impedance selector switches, located adjacent to the video input connectors. If the monitor is used in a loop-through operation, these switches should be set to the high impedence position. For single unit operation, or when the monitor is the last unit in a loop-through string, these switches should be set to the 75 ohm position to terminate the video cable.

**Table 2-1**
**REAR PANEL CONNECTORS**

| CONNECTOR | FUNCTION | CONNECTOR TYPE |
|---|---|---|
| J2 | 115VAC, 220VAC power input | IEC power cord |
| J3 | Red video input | BNC |
| J4 | Red video output | BNC |
| J5 | Green video with composite sync input | BNC |
| J6 | Green video with composite sync output | BNC |
| J7 | Blue video input | BNC |
| J8 | Blue video output | BNC |
| J9 | External sync input | BNC |
| J10 | External sync output<br>NOTE: For Ext. sync operations, S1 (located on the video PCB) should be switched to "Ext." | BNC |




HORIZONTAL TIMING




VERTICAL TIMING

06-5358

Figure 2-l
SYNC TIMING CHART

## 2.5 AC INPUT POWER

Before connecting the monitor to the power source, perform the following steps: (To remove the cover, refer to section 2.3.)

1. Determine that the line voltage and frequency are proper (115VAC ± 10%, 50/60 Hz).

2. Check the position of the input voltage plug on the power supply card. If the line voltage being used is not 115VAC, make certain that the plug is in the proper position for the input line voltage desired (JP7A - 115 VAC; JP7B - 220 VAC).

3. Plug the ac line cord into the power receptacle located on the rear connector panel.

## 2.6 INSTALLATION CHECKOUT

Proceed with the installation checkout as follows:

1. Apply a test pattern or some other suitable video to the video inputs.

2. Connect the monitor to an ac power source and set power switch S2 to the ON position. (There is normally a 15 second delay from application of power to the appearance of an image.)

3. Adjust the front panel brightness control for the desired viewing level.

4. If necessary, adjust the Horizontal Hold (located on the PCB video) and the Vertical Hold (located on the PCB deflection).

5. If color purity appears slightly out of adjustment, depress the Degauss switch (S1) and hold it down for 10 seconds, then release it.

6. If the monitor does not display an image, or if the displayed image is maladjusted and cannot be corrected, refer to Section 4 for maintenance and alignment procedures.

# SECTION 3

# OPERATION

## 3.1 INTRODUCTION

This section describes the procedures that should be followed when operating the 8830 Analog Color Monitor.

## 3.2 PREOPERATIONAL SETUP

Before operating the monitor, perform the following procedures:

1. Apply video input signals to the monitor, as specified in the Section 2 installation procedures.
2. Check that the input voltage selection plug on the power supply card is in the proper position for the applied line voltage (see section 2.5).
3. If it is desired to change the scanning rates or the video input levels, refer to Section 4 for setup procedures.

**NOTE**

Units are factory adjusted to operate at either an 18, 21, or 25 kHz horizontal rate at 50 Hz non-interlaced or to customer supplied formats.

## 3.3 OPERATION

### 3.3.1 Front Panel Controls

To operate, perform the following procedures:

1. Apply ac power to the monitor and depress the ac power switch to the ON position.
2. Allow approximately 2 minutes for warm-up.
3. Adjust the brightness control until the desired viewing level is achieved.
4. If color purity appears to be slightly out of adjustment, depress the Degauss switch (S1) and hold it down for 10 seconds, then release it.
5. If the display image is not satisfactory, refer to Section 4.

# SECTION 4

# MAINTENANCE

## 4.1 GENERAL

This section provides preventive and corrective maintenance procedures for the AYDIN CONTROLS Model 8830 Analog Color Monitor. Refer to section 1.5 for a list of recommended test equipment and special tools. While maintaining the monitor, be sure to strictly observe the safety precautions listed below.

***************
*** WARNING ***
***************

- OPERATING THIS MONITOR OUTSIDE THE CABINET OR WITH THE COVER REMOVED INVOLVES A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN FROM THE MONITOR POWER SUPPLIES, ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE PRECAUTIONS NECESSARY WHEN WORKING AROUND HIGH VOLTAGE EQUIPMENT SHOULD INSTALL OR MAINTAIN THIS EQUIPMENT.

a DO NOT INSTALL, REMOVE, OR HANDLE THE PICTURE TUBE IN ANY MANNER UNLESS SHATTER-PROOF GOGGLES ARE WORN. WHILE HANDLING THE PICTURE TUBE, BE SURE TO KEEP AWAY ANYONE NOT WEARING SHATTER-PROOF GOGGLES.

. A LINE ISOLATION TRANSFORMER IS REQUIRED WHEN TAKING MEASUREMENTS ON THE PRIMARY SIDE OF T702.

- X-RAY RADIATION WARNING: DO NOT READJUST POTENTIOMETERS SEALED AT THE FACTORY. THESE POTENTIOMETERS AFFECT THE X-RADIATION CHARACTERISTICS OF THIS PRODUCT.

- PRODUCT SAFETY NOTICE: MANY ELECTRICAL PARTS IN THIS MONITOR INCLUDE SPECIAL SAFETY RELATED CHARACTERISTICS. OFTEN THESE SAFETY CHARACTERISTICS CANNOT BE SEEN, NOR CAN REPLACEMENT PARTS OTHER THAN THE EXACT DESIGNATED PARTS PROVIDE THE SAME PROTECTION. ELECTRICAL COMPONENTS WITH SUCH SAFETY FEATURES ARE IDENTIFIED BY SHADING ON THE SCHEMATIC DIAGRAM.

**NOTE**

FCC: "THIS EQUIPMENT HAS NOT BEEN TESTED TO SHOW COMPLIANCE WITH NEW FCC RULES (47CFR PART 15) DESIGNED TO LIMIT INTERFERENCE TO RADIO AND TV RECEPTION. OPERATION OF THIS EQUIPMENT IN A RESIDENTIAL AREA IS LIKELY TO CAUSE UNACCEPTABLE INTERFERENCE TO RADIO COMMUNICATION, REQUIRING THE OPERATOR TO TAKE WHATEVER STEPS ARE NECESSARY TO CORRECT THE INTERFERENCE."

## 4.2 PREVENTIVE MAINTENANCE

Preventive maintenance for the 8830 Analog Color Monitor consists primarily of basic housekeeping tasks. Check all connections for proper fit, corrosion, sharply bent cables, exposed wires, and broken plugs. Periodically turn OFF all power, then air clean the chassis interior. Wipe off the chassis exterior and the CRT face with a lint-free cloth dampened with a mild cleaning solution. The cleaning frequency is determined by the environmental control at the user's site.

## 4.3 CORRECTIVE MAINTENANCE

Detailed alignment procedures, included for all internal adjustments, are to be performed following monitor repair and periodic alignment.

### 4.3.1 Alignment Procedures

This section contains monitor alignment procedures that should be performed after the replacement of a major component, such as a CRT or circuit board. Degaussing should be performed periodically and whenever the monitor is moved from one location to another.

For quick reference, Table 4-l lists all maintenance adjustments. Component locator diagrams (Figures 4-3 through 4-6) show the location of all adjustments.

In the following alignment procedures, it is assumed that proper line voltage and frequency are available. Check that connector JP7A or JP7B (on the power supply board) is in the desired position for 115 VAC or 220 VAC.

The red, green, and blue inputs require application of a video source with proper line rate. The green video must contain composite sync or a combined horizontal and vertical signal supplied to the external sync input. If external sync is desired, set switch S1 (located on the video PCB) to the "External Sync" position. Return S1 to "Internal Sync" when using composite sync on green.

The video source should be a cross-hatch or dot pattern for static convergence alignment.

After connecting all inputs, allow approximately 15 minutes for warm-up before proceeding.

Table 4-l
MAINTENANCE ADJUSTMENTS CROSS REFERENCE

| FUNCTION | LOCATION | DESIGNATION | SECTION |
|---|---|---|---|
| Ac ON/OFF switch | Front panel | s2 | ----- |
| Red terminator switch | Rear panel | S3 | 4.3.1 |
| Green terminator switch | Rear panel | s4 | 4.3.1 |
| Blue terminator switch | Rear panel | s5 | 4.3.1 |
| Sync terminator switch | Rear panel | S6 | 4.3.1 |
| Sync switch | Video PCB | Sl | 413.1 |
| Degauss control | Front panel | Sl | 4.4 |
| Brightness control | Front panel | VR902 | 4.5.1, step 1 |
| Horizontal hold | Video PCB | R33 | 4.5.1. step 2 |
| Vertical hold | Deflection PCB | R502 | 4.5.1, step 3 |
| Width | Deflection PCB | R602 | 4.5.2, step 1 |
| Heisht | Deflection PCB | R505 | 4.5.2, step 2 |
| G2 | Deflection PCB | R638 | 4.5.2, step 4 |
| Horizontal phase | Video PCB | R23 | 4.5.2, step 5 |
| Vertical centering | Deflection PCB | R526 | 4.5.2, step 7 |
| Horizontal centering | Deflection PCB | R631 | 4.5.2, step 7 |
| Side pincushion | Deflection PCB | R520 | 4.5.2, step 8 |
| Vertical linearity | Deflection PCB | R507 | 4.5.2, step 9 |
| Focus control | Rear panel | VR901 | 4.5.2, step 10 |
| Red bias control | Video PCB | R235 | 4.5.3, step 3 |
| Green bias control | Video PCB | R135 | 4.5.3, step 3 |
| Blue bias control | Video PCB | R335 | 4.5.3, step 3 |
| Red gain control | Video PCB | R210 | 4.5.3, step 5 |
| Green gain control | Video PCB | R110 | 4.5.3, step 5 |
| Blue gain control | Video PCB | R310 | 4.5.3, step 5 |

## 4.4 DEGAUSSING

Poor color purity or poor convergence indicates the need for degaussing. Degaussing should precede any other attempt to adjust color purity or convergence.

The display monitor contains a built-in picture tube degaussing coil. To degauss the picture tube, depress the Degauss switch (Sl) on the front panel and hold it down for approximately 10 seconds to ensure proper operation of the degauss circuit.

Other parts of the monitor may also require degaussing. This is indicated by poor color purity or static convergence which cannot be corrected by normal alignment.

The monitor chassis is degaussed by manually using a commercial degaussing coil as follows:

1. With the coil switch in the OFF position, and with the degaussing coil 6 to 8 feet from and perpendicular to the screen, turn the switch to the ON position.

2. Turn the coil parallel to the screen, and in a circular motion, slowly bring the coil to the monitor.

3. Continuing the circular motion, pass the coil over the front, top, and sides of the monitor for approximately 2 minutes.

4. Turn the coil perpendicular to the monitor, and in a circular motion, slowly back away 6 to 8 feet, then turn the coil switch OFF.

## 4.5 SET-UP ADJUSTMENT PROCEDURE

Be sure to follow this procedure in sequence and only make adjustments when required. Use an insulated screwdriver for the adjustments. Apply a known, all white test pattern signal, such as a crosshatch or full page of "E"s.

**NOTE**

If there is no sync signal present on the green video input or the external input, there will be no visible raster present. If this condition exists, check for the following:

1. Correct setting of the INT/EXT sync switch (S1) located on the video PCB.

2. If item "1" is correct, connect a scope to the sync line and check for proper signal amplitude.

### 4.5.1 Horizontal and Vertical Synchronization

1. Turn front panel brightness control fully clockwise.

2. Adjust horizontal hold R33 (PCB Video) until the picture locks in the horizontal direction.

3. Adjust vertical hold R502 (PCB DEFL) until the picture locks in the vertical direction.

4. To ensure that the vertical and horizontal holds have been adjusted accurately with R33 and R502, switch the INT/EXT switch S1 (PCB Video) between the internal and external mode. This will ensure that the controls were not set at the edge of their lock range.

### 4.5.2 Raster Size, Centering, and Geometric Adjustments

1. Adjust width control R602 for the desired picture width.
2. Adjust height control R505 for the desired height.
3. Turn the front panel brightness control counterclockwise until a dim test pattern display is present.
4. Adjust G2 control R638 to obtain a dim display of the background raster.
5. Adjust horizontal phase control R23 (PCB video) until the active video is centered on the background raster.
6. Adjust G2 control R638 until the background raster just disappears.
7. Adjust horizontal and vertical centering controls R631 and R526 until the active video is centered on the screen.
8. Adjust side pincushion control R520 for desired straightness of raster sides.
9. Adjust vertical linearity control R507 for best overall vertical linearity. (If necessary, readjust the vertical centering and vertical linearity.1
10. Adjust focus VR901 to obtain optimum line sharpness.

### 4.5.3 Input Signal Level Compensation and Gray Scale Tracking

The signal level compensation and gray scale tracking must be made with no more than 30% video present on the raster.

**NOTE**

A four-inch, white video square block in the center of the screen is ideal for set-up, because this will ensure proper operation of the beam limiter circuit.

1. Depress the Degauss switch (S1) on the front panel for approximately 10 seconds, then release it.
2. Turn the front panel brightness control until the video display is just visible.
3. Adjust red, green, and blue bias controls R235, R135, and R335 for a gray display. Only adjust the control color that is needed to make gray. If the background raster appears, readjust G2 control R638.

4. Adjust the front panel brightness control fully clockwise.

5. Adjust red, green, and blue gain controls R210, R110, and R310 for a white non-blooming display.

6. Turn the front panel brightness control up and down slowly and check for gray scale tracking. Repeat steps 2 through 5 to obtain good gray scale tracking,

7. Adjust the rear panel focus control for the best overall crispness of display.

### 4.5.4 Purity and Convergence Procedure

The monitor should not require purity and convergence adjustments with time, as do some delta gun CRT systems. This is because the monitor's convergence does not depend on external dynamic circuits, nor do changes in sweep or high voltage greatly affect it.

Adjustments should be necessary only if the CRT deflection yoke or ring assembly has been disturbed. (Refer to Figure 4-1.)

**NOTE**

Before making any adjustments, degauss the monitor as outlined in section 4.4. If any adjustments do become necessary, be extremely careful not to touch exposed components or high voltage terminals,

#### 4.5.4.1 Purity Adjustment

Apply an input signal to yield a full green field. Observe the variation in light intensity and color. If the picture *is* not completely uniform with green, loosen the lock ring and adjust the purity magnet for best overall uniformity.

**NOTE**

Take care not to disturb the convergence magnets.

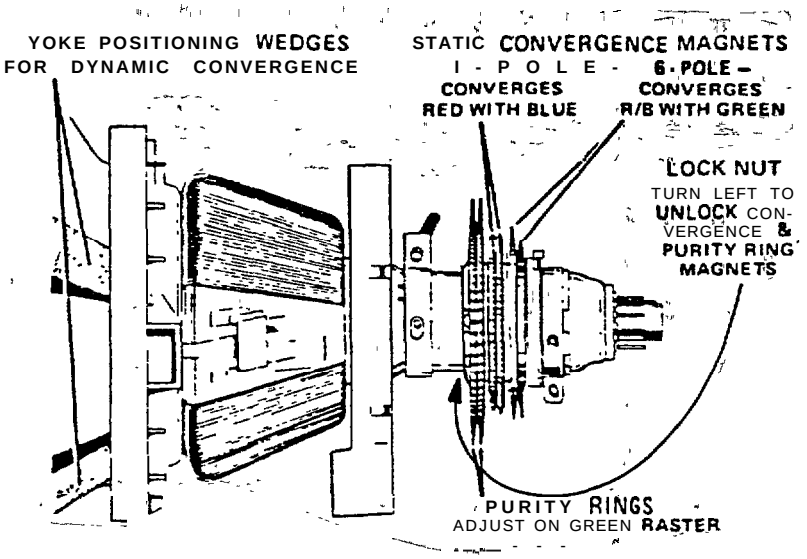



**Figure 4-1**
**PURITY AND CONVERGENCE ADJUSTMENTS**

#### 4.5.4.2 Convergence Adjustments

The monitor's convergence adjustments are very sensitive, and several locations may be found that will yield the same convergence correction. Therefore, it is not necessary to move the adjustments more than required to correct the misconvergence. The final setting should be very close to the original setting, which is marked by a painted line across the magnet rings.

Program a cross-hatch picture. Examine convergence throughout the picture for uniformity.

**NOTE**

Do not try to adjust for misconvergence if the error exists only in one area of the tube. Adjustment of rings compensates uniformly across the tube. Some small error will exist after rings are properly adjusted. Refer to the misconvergence tolerances in Table l-l under "Performance" characteristics.

If convergence adjustment is required, proceed as follows:

1. Adjust brightness to mid-scale.

2. Adjust for best focus.

3. Loosen the lock ring.

4. Adjust the second set of rings for red and blue misconvergence, as shown in Figure 4-2. Adjusting these tabs converges the red and blue beams independently of the green.

5. Adjust the third set of rings for green onto red and blue with beam directions as indicated by Figure 4-2.

**NOTE**

If misconvergences are substantial, it may be necessary to repeat adjustments between the second and third ring set.

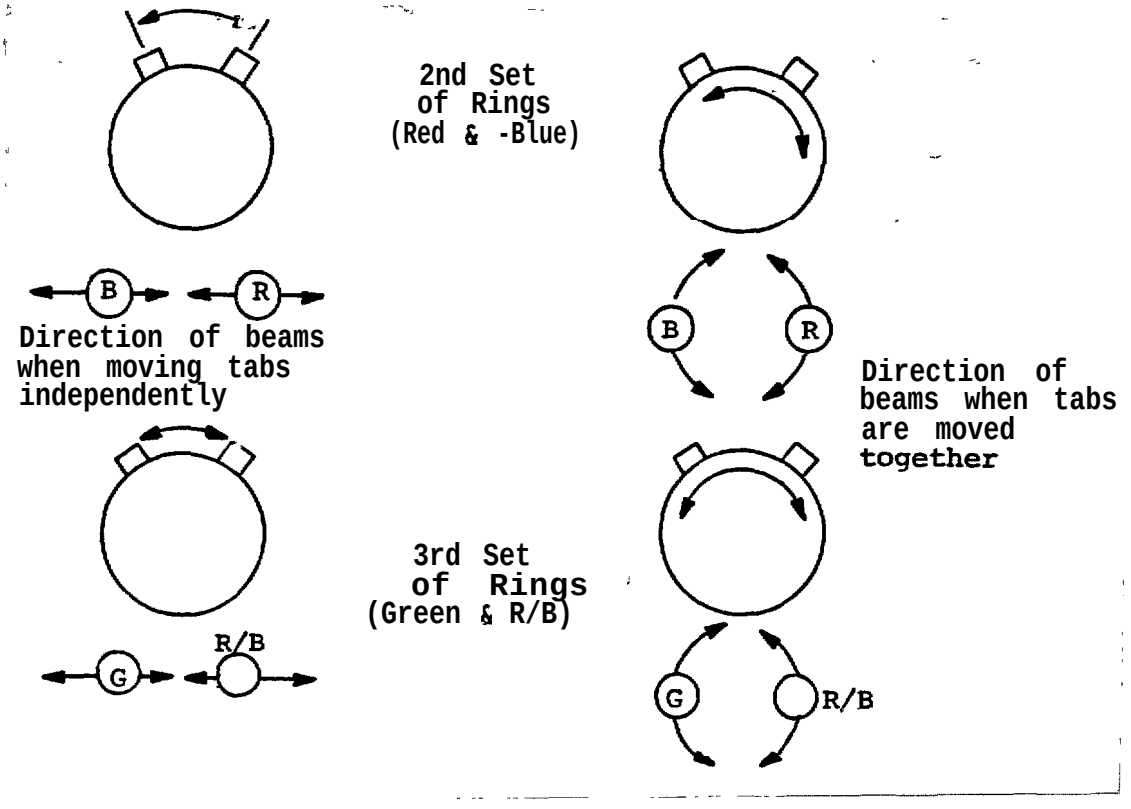



Figure 4-2
RING ADJUSTING

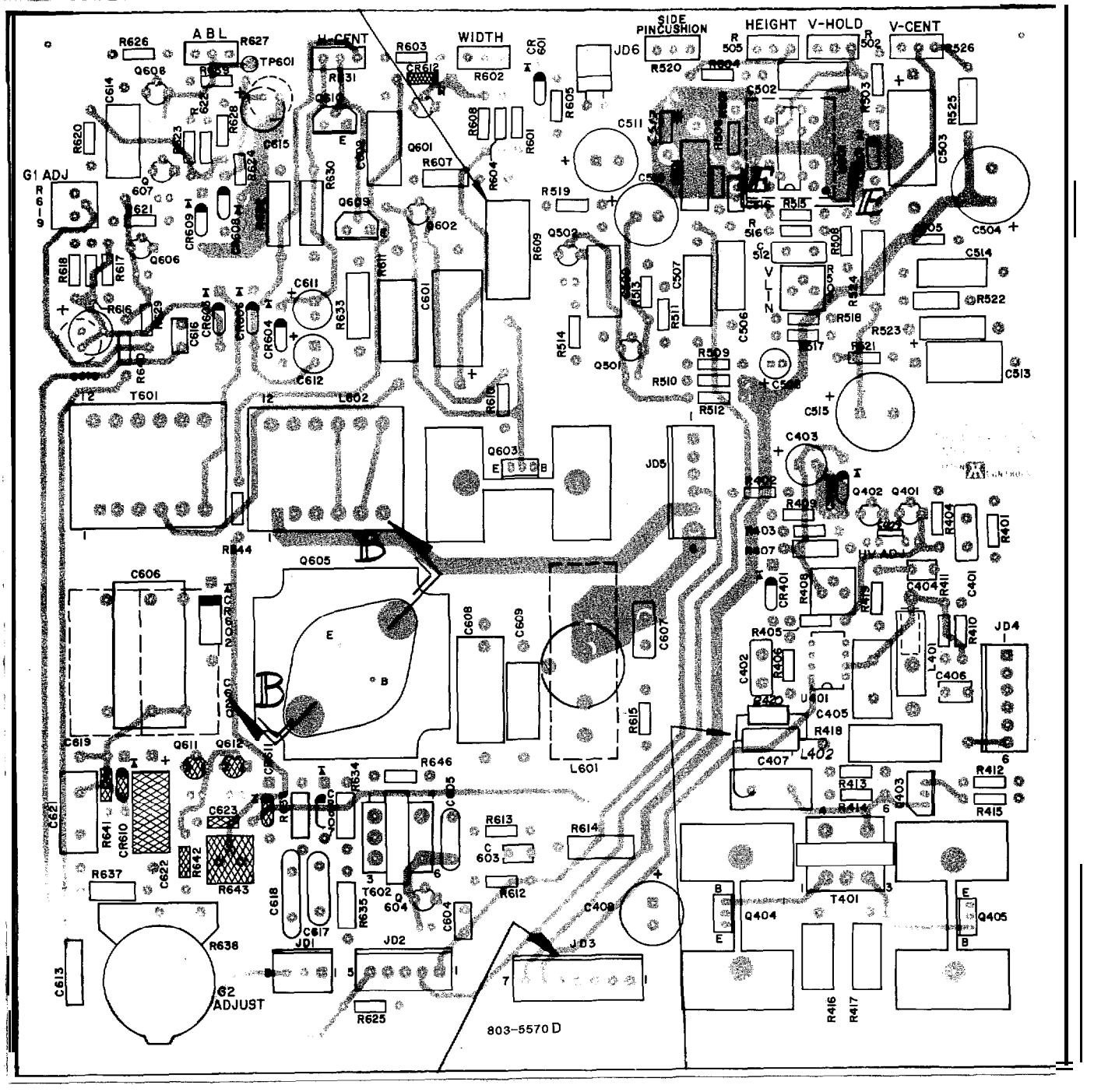



Figure 4-3
COMPONENT LOCATOR DIAGRAM
DEFLECTION CARD ASSEMBLY (DFL-003)

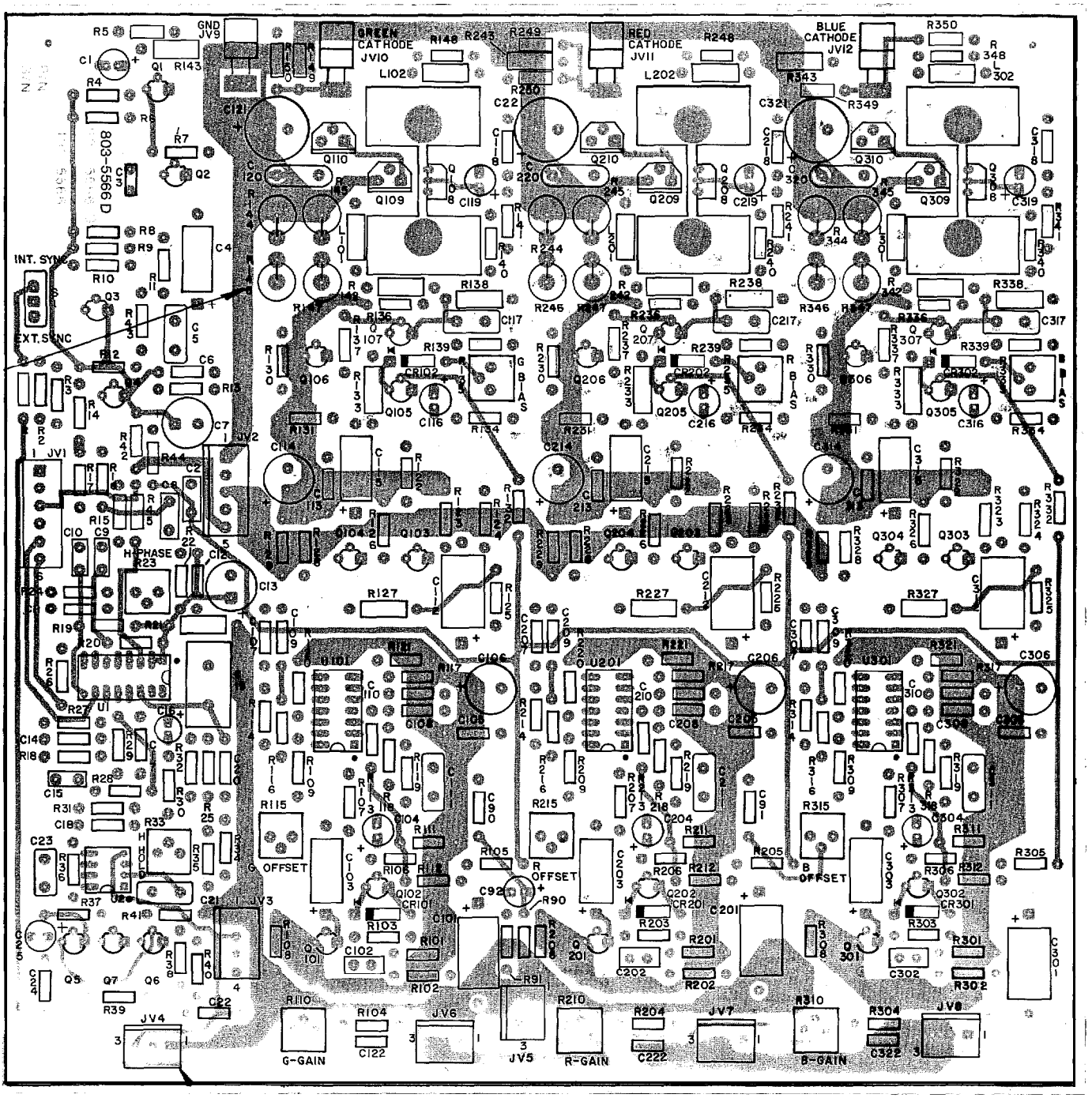



Figure 4-4
COMPONENT LOCATOR DIAGRAM
VIDEO CARD ASSEMBLY (VID-122)

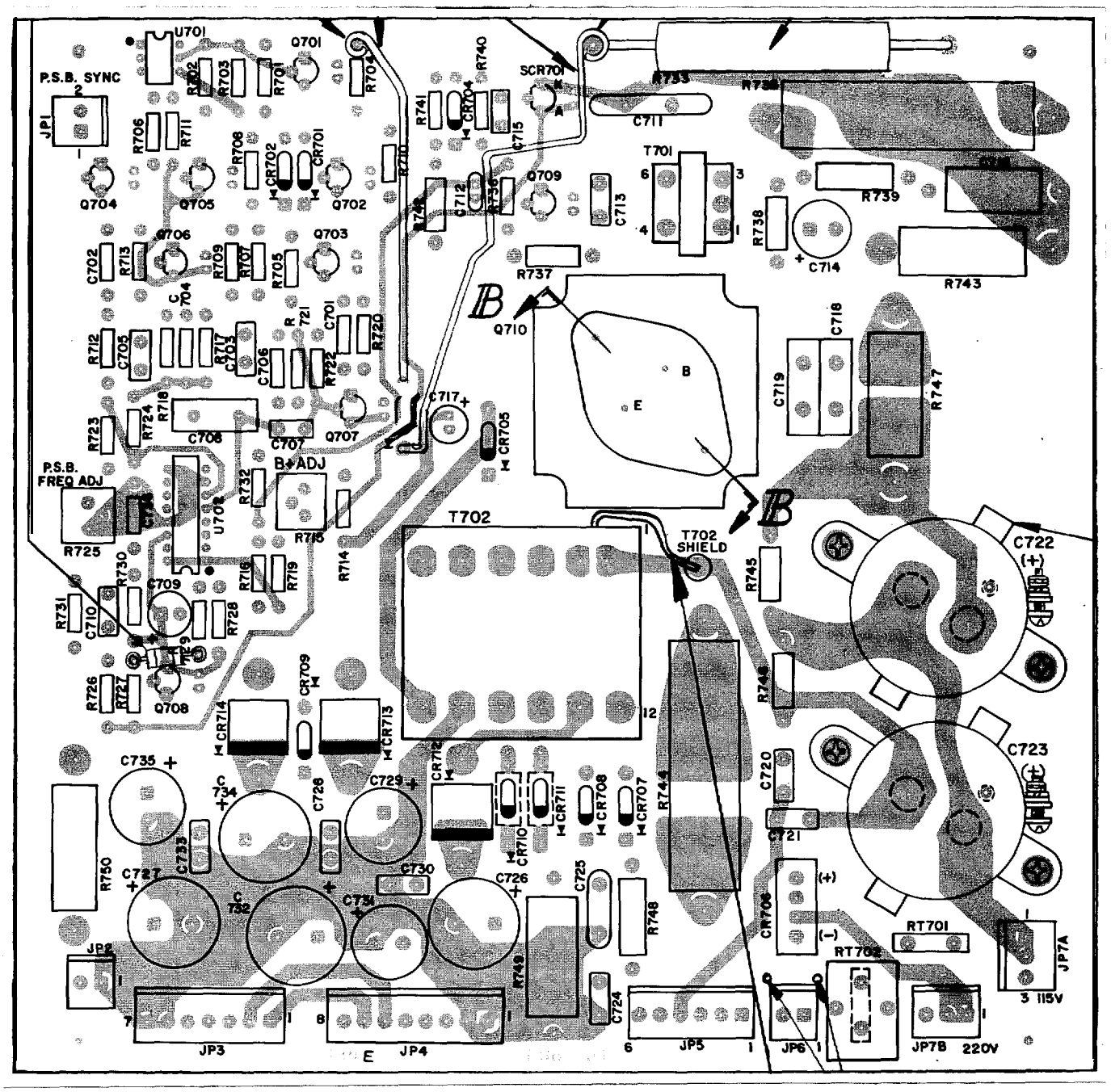



Figure 4-5
COMPONENT LOCATOR DIAGRAM
POWER SUPPLY CARD ASSEMBLY (PSB-001)

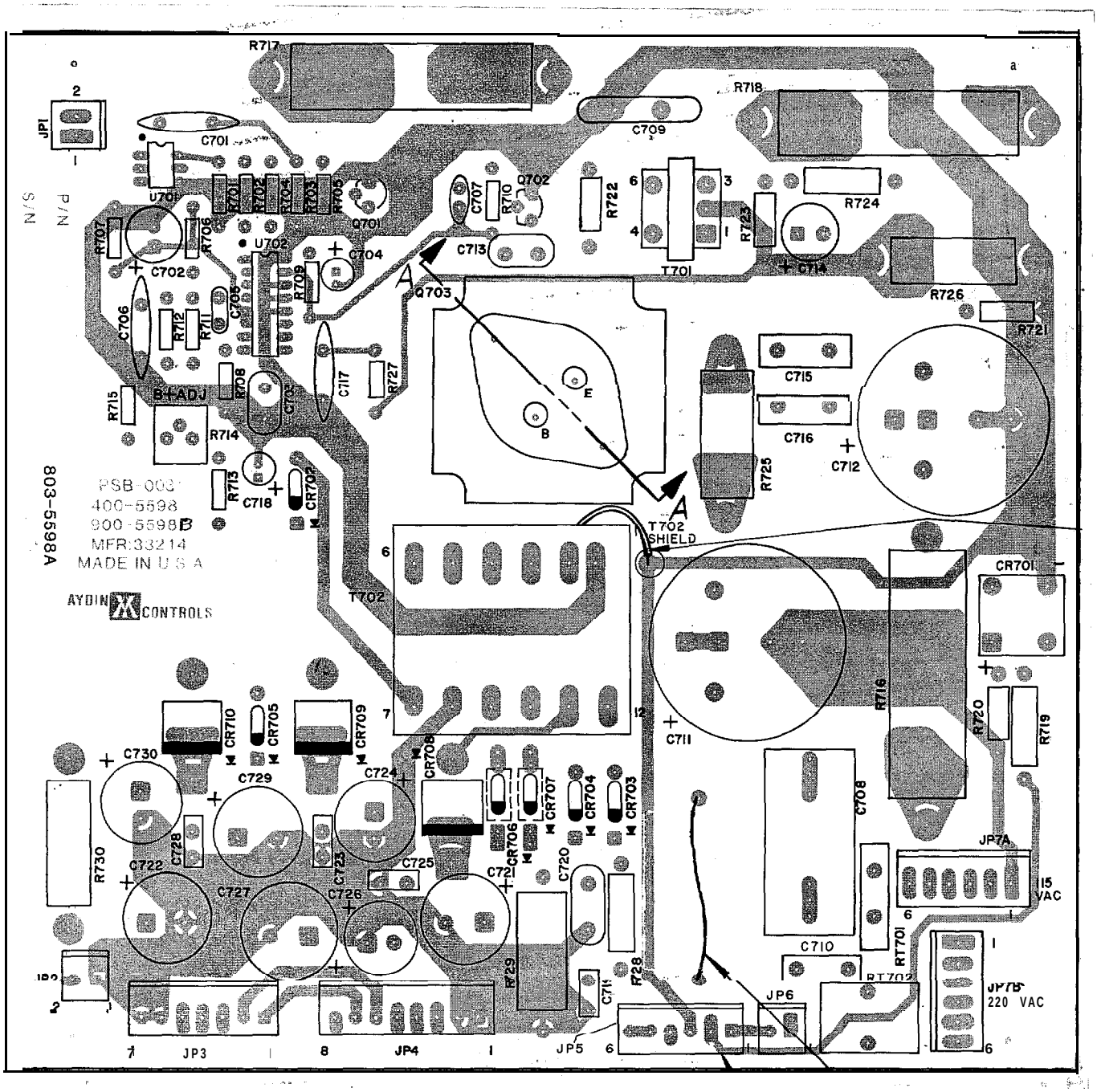



Figure 4-6
COMPONENT LOCATOR DIAGRAM
POWER SUPPLY CARD ASSEMBLY (PSB-003)

# SECTION 5

# PARTS

## 5.1 INTRODUCTION

Table 5-1 contains a list of replaceable subassemblies and chassis piece parts. The part numbers listed below are for reference only. Please consult the factory for specific part numbers and revision levels.

Table 5-l
SPARE SUBASSEMBLIES AND CHASSIS PARTS

| AYDIN PART NO. | DESCRIPTION |
|---|---|
| 460-5486-501 | PSB001 card assembly (15-18.5 kHz) |
| 460-5486-502 | PSB001 card assembly (18.5-21.5 kHz) |
| 460-5486-503 | PSB001 card assembly (21.5-25 kHz) |
| 460-5598-501 | PSB003 card assembly (15-18.5 kHz) |
| 460-5598-502 | PSB003 card assembly (18.5-21.5 kHz) |
| 460-5598-503 | PSB003 card assembly (21.5-25 kHz) |
| 460-5566-501 | VID122 card assembly (15-18.5 kHz) |
| 460-5566-502 | VID122 card assembly (18.5-21.5 kHz) |
| 460-5566-503 | VID122 card assembly (21.5-25 kHz) |
| 460-5570-501 | DFL003 card assembly (15-18.5 kHz) |
| 460-5570-502 | DFL003 card assembly (18.5-21.5 kHz) |
| 460-5570-503 | DFL003 card assembly (21.5-25 kHz) |
| *101-188-001 | *Focus pot assembly |
| 101-189-001 | Gain pot |
| *101-190-001 | Focus pot |
| *101-190-002 | Focus pot |
| 105-025-001 | CRT and yoke assembly, 19V, standard phosphor |
| 105-025-002 | CRT and yoke assembly, 19V, long persistence phosphor |
| 107-260-002 | Termination switches, S3-S6 |
| 109-118-001 | High voltage transformer, T901 (15-18.5 kHz) |
| 109-118-002 | High voltage transformer, T901 (18.5-21.5 kHz) |
| 109-118-003 | High voltage transformer, T901 (21.5-25 kHz) |
| 110-044-002 | Indicator lamp |
| 111-006-020 | Fuse, 2A, 250V, 3AG (For 220V operation only) |
| 111-006-022 | Fuse, 3A, 250V, 3AG (For 115V operation only) |
| 135-316-001 | Capacitor module, M902 |
| *136-212-001 | *High voltage multiplier, M901 |
| *136-212-002 | *High voltage multiplier, M901 |
| Consult factory | Rocker switch, Degauss, S1 |
| Consult factory | Rocker switch, Power, S2 |

* Components that are matched sets and should be replaced with matched components.

**GENERAL ELECTRIC COMPANY . DRIVE SYSTEMS DEPARTMENT**
**SALEM, VA. 24153**

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